Blood related drugs, Muscle relaxant & Analgesics



Talent Management Department

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United Pharmacy

Today we will learn about

01 Blood Related Drugs

02 Muscle relaxant

03 Analgesics and Antipyretics

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Blood Related drugs classification



Drugs that increase flowability

1- Anti-platelet Agents

- Inhibit platelet aggregation, the early step in clot formation. They're mainly used in arterial thrombotic conditions (e.g. heart attacks, strokes).

Classes:

A. COX Inhibitors (e.g. Aspirin): ✅

- Irreversibly inhibits COX-1 → ↓ Thromboxane A2 → ↓ platelet aggregation

B. $P2Y_{12}$ Inhibitors (e.g. Clopidogrel, Prasugrel, Ticagrelor): ✅

- Block ADP receptor ($P2Y_{12}$) on platelets → ↓ activation & aggregation.
- **Uses**: Acute coronary syndrome, Ischemic stroke, Post-stent placement, Preventing clot recurrence in high-risk patients.

- **Drugs that increase flowability**
  - 1 **Anti-Platelets**
    - COX inhibitors
      - ▪ **ASPIRIN PROTECT®**| *Acetyl salicylic acid*
    - **Anti-Platelets**
      - ADP receptor ($P2Y_{12}$) antagonists
        - ▪ **PLAVIX®** | *Clopidogrel*
        - ▪ **BRILINTA ®** | *Ticagrelor*
        - ▪ **PEVAS®** | *Prasugrel*
  - 2 **Anti-Coagulants**

Drugs that increase flowability

2. Anti-coagulants

- Prevent formation and extension of fibrin clots, mainly used in venous and cardiac thrombosis.

Classes & MOA:

A. Vitamin K Antagonist: ✅

- Inhibits vitamin K epoxide reductase → ↓ synthesis of clotting factors II, VII, IX, X

B. Direct Thrombin Inhibitors: ✅

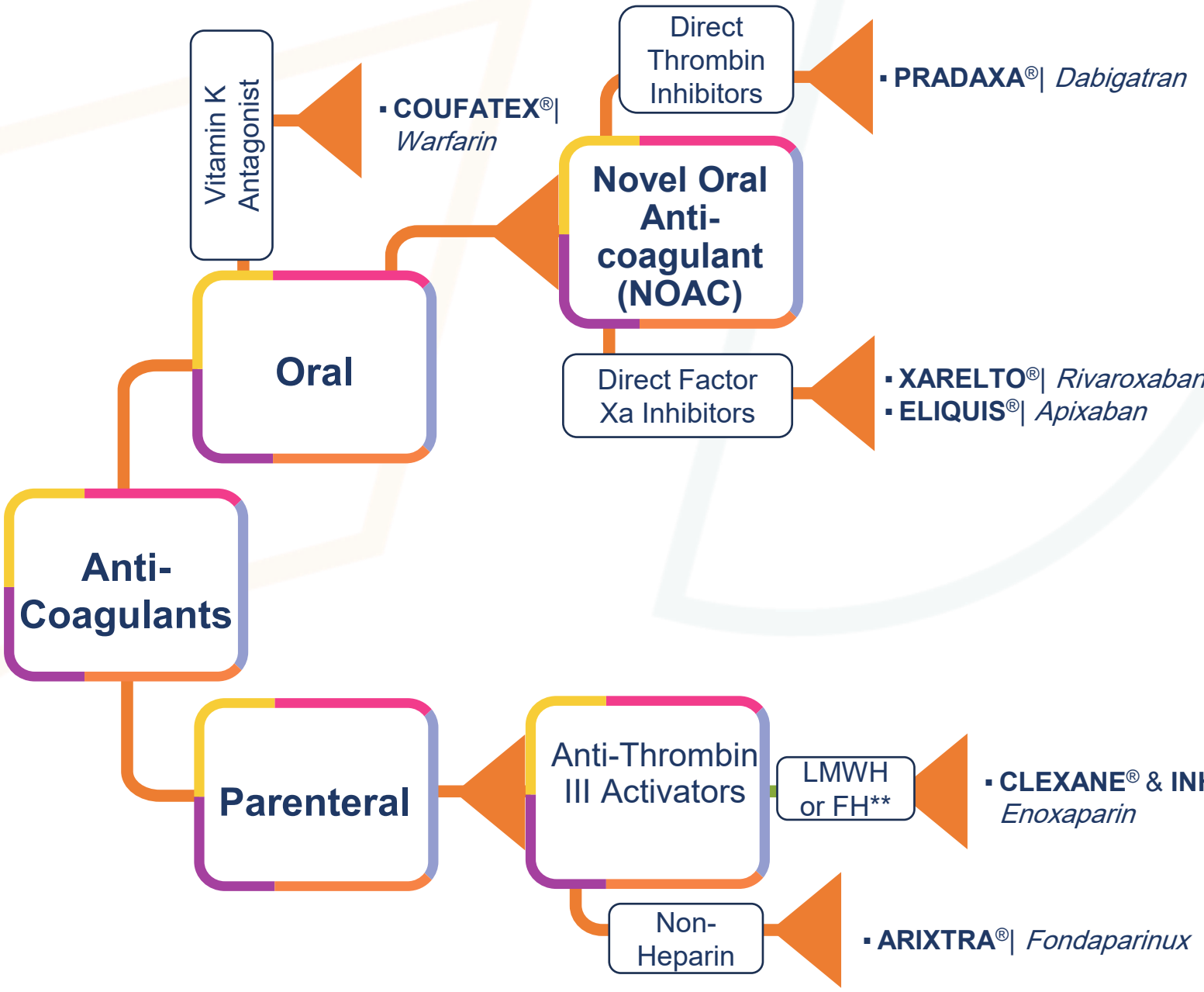
- Directly inhibit thrombin (Factor IIa) → prevents fibrin formation.

C. Direct Factor X Inhibitors: ✅

- Inhibit Factor Xa directly → prevent thrombin generation.

D. Antithrombin III Activators (Injectables) ✅

- These enhance the activity of antithrombin III, which inactivates several clotting factors (esp. thrombin and Factor Xa).



** LMWH= Low Molecular Weight Heparin.
** FH= Fractionated Heparin.

Drugs that decrease flowability

These drugs prevent breakdown of clots by inhibiting plasmin, useful in controlling excessive bleeding.

Uses:

- Heavy menstrual bleeding,
- Bleeding during surgeries (e.g. cardiac, dental)
- Trauma or postpartum hemorrhage, Hemophilia (as adjunct)

Hemopoietic agents

These drugs stimulate blood cell production to treat or prevent anemia and thrombocytopenia, especially in chemotherapy, renal failure, or bone marrow suppression.

1. Thrombopoiesis-Stimulating Agents: ✅

- Stimulate platelet production by activating thrombopoietin receptors.

2. Erythropoiesis-Stimulating Agents (ESAs) ✅

- Stimulate red blood cell production by mimicking endogenous erythropoietin.
- Binds to **EPO** receptors → ↑ RBC production.

Drugs that decrease flowability

- Anti-Fibrinolytics
  - **LYSTEDA®** | Tranexamic acid
  - **CYKLOKAPRON ®** | Tranexamic acid
  - **DICYNONE®** | Ethamsylate

Hemopoietic agents

- Thrombopoiesis “Drugs ↑ platelets”
  - Oral
    - **REVOLADE®**| Eltrombopag
  - Subcutaneous (SC)
    - **NPLATE®**| Romiplostim
- Erythropoiesis “Drugs ↑ RBC’s”
  - **ARANESP®**| Darbepoetin α

Muscle relaxant drugs classification



Muscle relaxant

Muscle relaxants are medications used to reduce muscle spasms, spasticity, or hypertonia caused by conditions such as muscle injury, neurological disorders, or surgery. They work by interfering with nerve signals, either centrally in the CNS or peripherally at the neuromuscular junction.

1. Centrally Acting Agents ✅

- These act on the central nervous system (brain or spinal cord) to reduce excessive muscle tone or spasms.
- **Note:** May cause drowsiness, dizziness, and CNS depression.

2. Peripherally Acting Agents ✅

- These act directly at the neuromuscular junction or within the muscle fibers, interfering with the contractile process.

Muscle relaxant drugs

Centrally Acting Agents

- **BACLON®** | Baclofen
- **RELAXON®** | Chlorzoxazone
- **BENZAFLEX®** | Cyclobenzaprine
- **TILAX®** | Tizanidine

Peripherally Acting Agents

- Dantrolene
- Botulinum Toxin
- Atracurium
- Cisatracurium

N.B: Theye are not currently available

Analgesics & antipyretics drugs classification



Analgesics & Antipyretics Drugs

Analgesics are medications used to relieve pain without causing loss of consciousness. They act by blocking pain signals and/or reducing inflammation.

1- Acetaminophen (Paracetamol): ✅

- Has an antipyretic & analgesic effects only (has no anti-inflammatory effect).
- Most safe so can be used in patients with gastric complaints/risks.
- Drug of choice for pregnant women & children with viral infections.

2- Nonsteroidal anti-inflammatory drugs (NSAIDs):

- NSAIDs act through inhibition of the synthesis of prostaglandins.
- Aspirin is contraindicated in children less than 16years old due to risk of Reye's syndrome.
- Can be classified into selective and nonselective.
- Selective COX-2 inhibitors are more suitable for patients with GIT problems.
- Nonselective COX inhibitors are suitable for patients with cardiac problems.

Analgesics & Antipyretics

Paracetamol

- **ADOL®**
- **FEVADOL ®**
- **EMIDOL®**

NSAIDs

Non-selective COX-1 & COX-2

- **SAPOFEN®** | Ibuprofen
- **EMIFENAC®** | Diclofenac Sodium
- **CATAFAST®** | Diclofenac Potassium
- **PROXEN®** | Naproxen
- **PONSTAN FORT®** | Mefenamic Acid
- **KETOFAN®** | Ketoprofen
- **MOBIC®** | Melxicam

Selective COX 2

- **CELEBREX®** | Celecoxib
- **ARCOXIA®** | Etoricoxib

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